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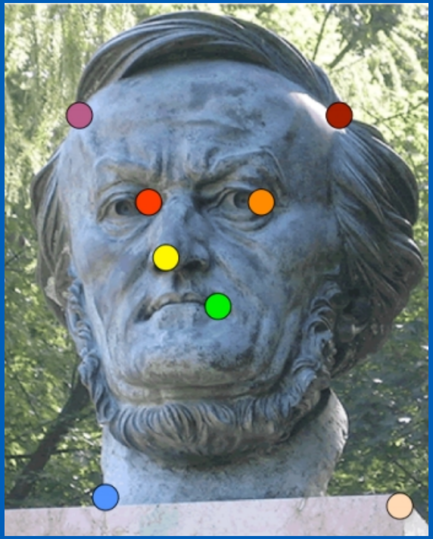
8 Points Algorithm

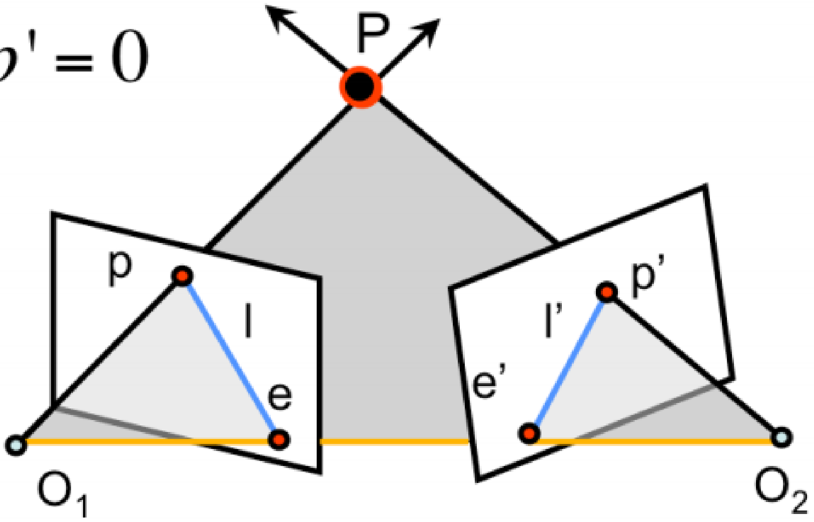


Summary

- Recap epipolar constraints
- 8 points method
- Normalized 8-points method

Epipolar constraints

$$p^T \cdot F\ p' = 0$$



- l = Fp' ⟶ epipolar line for p
- l' = F^Tp ⟶ epipolar line for p'
- F is a 3x3 matrix with 7 DOF
- F is singular and its rank is 2

Fundamental Matrix

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- It is not necessary to fully know their calibration parameters of a stereo vision system
- A reduced set of relations between the two views suffices
- The fundamental matrix F encodes this set
  - F can be used to relate the projections of the same world point P on both images (p & p')
  - F can be used to determine epipolar lines then easing the search of correspondent points in both images
  - It can be also used to rectify stereo images

Fundamental Matrix estimation

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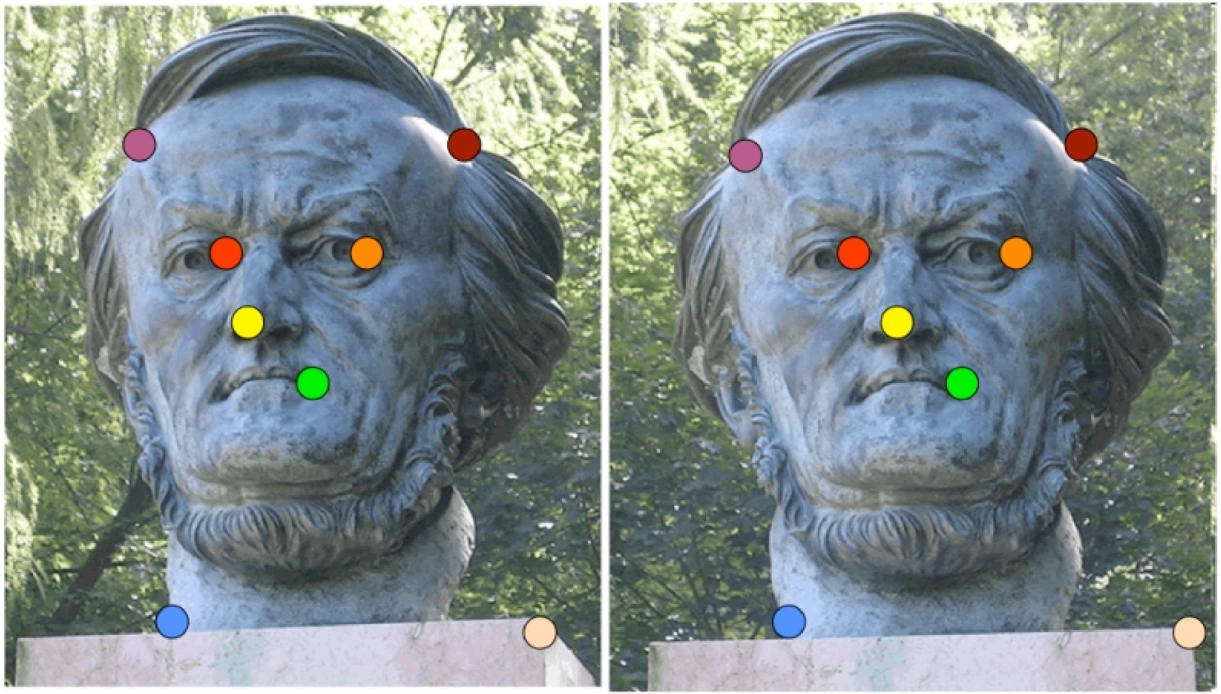

- Can we estimate F without knowing <u>nothing</u> about K, R, T?
- Yes → 8-point algorithm

Fundamental Matrix Estimation

- Problem:
  - Given a set of matched image points in both images $\{p_i, p'_i\}$
  - Estimate the Fundamental Matrix F
- How many matches do we need?
- How we perform the estimation?

Expand equation

- Each correspondence should satisfy the epipolar constraint

$$p_i^T F p_i' = 0$$

$$p_i = \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} \qquad p_i' = \begin{bmatrix} u_i' \\ v_i' \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} u_i & v_i & 1 \end{bmatrix} \begin{bmatrix} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{bmatrix} \begin{bmatrix} u_i' \\ v_i' \\ 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} u_i u_i' & u_i v_i' & u_i & v_i u_i' & v_i v_i' & v_i & u_i' & v_i' & 1 \end{bmatrix} \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} = 0$$

Linear Homogeneous System

- One correspondence ⟶ One equation
- Multiple correspondences ⟶ **Linear Homogenoeous System**

$$\begin{bmatrix} u_1u_1' & u_1v_1' & u_1 & v_1u_1' & v_1v_1' & v_1 & u_1' & v_1' & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_iu_i' & u_iv_i' & u_i & v_iu_i' & v_iv_i' & v_i & u_i' & v_i' & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_nu_n' & u_nv_n' & u_n & v_nu_n' & v_nv_n' & v_n & u_n' & v_n' & 1 \end{bmatrix} \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} = 0$$

Correspondences

- Therefore 1 correspondence $\longrightarrow$ 1 equation
- 9 unknowns $\longrightarrow$ 9 equations?
  - No, fundamental matrix has 7 DOF
- Therefore 7 correspondences should be enough
  - Yes, if we would like to explore complex non linear methods
  - We definitely would opt for a fast linear solution
  - Then use 8 points!

Correspondences

- Take then, at least, 8 correspondences

$$\begin{bmatrix} u_1u'_1 & u_1v'_1 & u_1 & v_1u'_1 & v_1v'_1 & v_1 & u'_1 & v'_1 & 1 \\ u_2u'_2 & u_2v'_2 & u_2 & v_2u'_2 & v_2v'_2 & v_2 & u'_2 & v'_2 & 1 \\ u_3u'_3 & u_3v'_3 & u_3 & v_3u'_3 & v_3v'_3 & v_3 & u'_3 & v'_3 & 1 \\ u_4u'_4 & u_4v'_4 & u_4 & v_4u'_4 & v_4v'_4 & v_4 & u'_4 & v'_4 & 1 \\ u_5u'_5 & u_5v'_5 & u_5 & v_5u'_5 & v_5v'_5 & v_5 & u'_5 & v'_5 & 1 \\ u_6u'_6 & u_6v'_6 & u_6 & v_6u'_6 & v_6v'_6 & v_6 & u'_6 & v'_6 & 1 \\ u_7u'_7 & u_7v'_7 & u_7 & v_7u'_7 & v_7v'_7 & v_7 & u'_7 & v'_7 & 1 \\ u_8u'_8 & u_8v'_8 & u_8 & v_8u'_8 & v_8v'_8 & v_8 & u'_8 & v'_8 & 1 \end{bmatrix} \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} = Af = 0$$

Fundamental Matrix Estimation

- Homogeneous system
- F_{ij}s are the unknowns
  - We can find them up to a scale $\longrightarrow$ impose norm as 1
- A not 0 solution is possible
  - For 8 points when rank of A is 8
  - Noise and quantization can lead to have 9 as rank (9 columns)

Fundamental Matrix Estimation

- Again, total least squares approach
  - $Af = 0 \longrightarrow$ minimize $||Af||^2$
- SVD can be used
  - In order to avoid the trivial zero solution and bypass scale issue impose the norm to be 1
  - $||F||^2 = 1$

Fundamental Matrix Estimation

- Find the SVD of $A^T A$
- Entries of f (or F) are the column of V corresponding to the least singular value

Fundamental Matrix Estimation

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- Solving previous system gives us a $\hat{F}$ solution that satisfies
  - $p^T\hat{F}p'=0$
- Anyway $\hat{F}$ is not necessarily a proper fundamental matrix
  - i.e. it can not be a rank 2 matrix
- Let's look for a best rank 2 approximation
  - We seek for an F that minimizes the Froebenius norm of $\hat{F}$
  - Subject to the constraint that $\det(F) = 0$

Fundamental Matrix Estimation

- Therefore we have to solve
  - $||F - \hat{F}|| = 0$ (Froebenius norm)
  - $\det(F) = 0$
- Froebenius norm of a matrix:
  - Sq. root of sum of squares of all entries of the matrix
- How to solve? $\longrightarrow$ SVD again

Fundamental Matrix Estimation

$$\|F-\hat{F}\|=0 \qquad det(F)=0$$

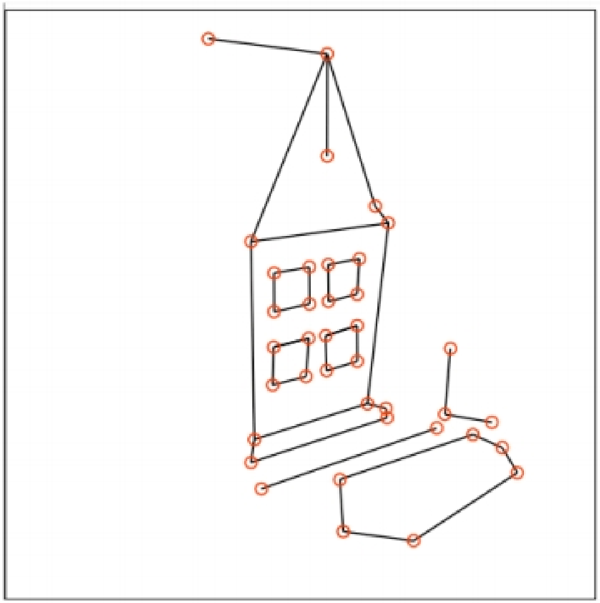
$$F=U\begin{bmatrix} s_1 & 0 & 0 \\ 0 & s_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}V^T \qquad UDV^T=SVD(\hat{F})$$

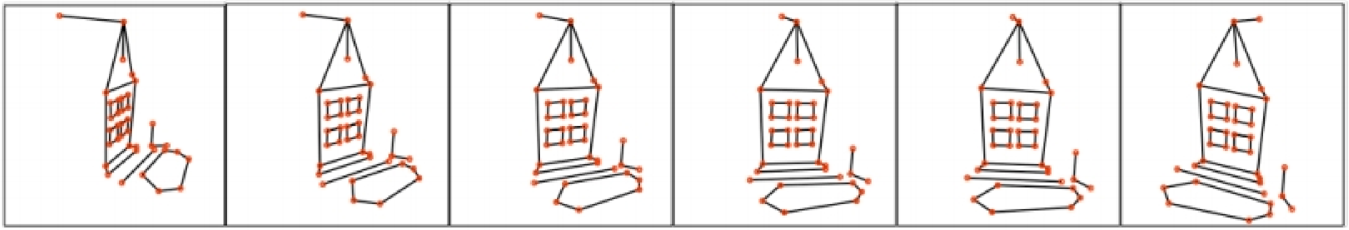
[HZ] pag 281, chapter 11, “Computation of F”

Fundamental Matrix Estimation

- The 8-point algorithm may be then formulated as consisting of two steps
  1) Linear solution. A solution $\hat{F}$ is obtained from the vector f corresponding to the smallest singular value of A
  2) Constraint enforcement. Replace $\hat{F}$ by F , the closest singular matrix to $\hat{F}$ under a Frobenius norm. This correction is done using the SVD
- The algorithm thus stated is extremely simple, and readily implemented

Slection of points





Data courtesy of R. Mohr and B. Boufama.

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Normalized 8 Points Algorithm

Normalized 8 points algorithm

- The 8-point algorithm features a numerical issue $$Af = 0$$
- Matrix A can be a not well conditioned (unbalanced) matrix
  - Values of matrix A are not necessarily in the same order of magnitude
  - This can badly affect the SVD decomposition
- No invariance wrt similarity ([SR T])

See HZ page 108 for details

Normalized 8 points algorithm

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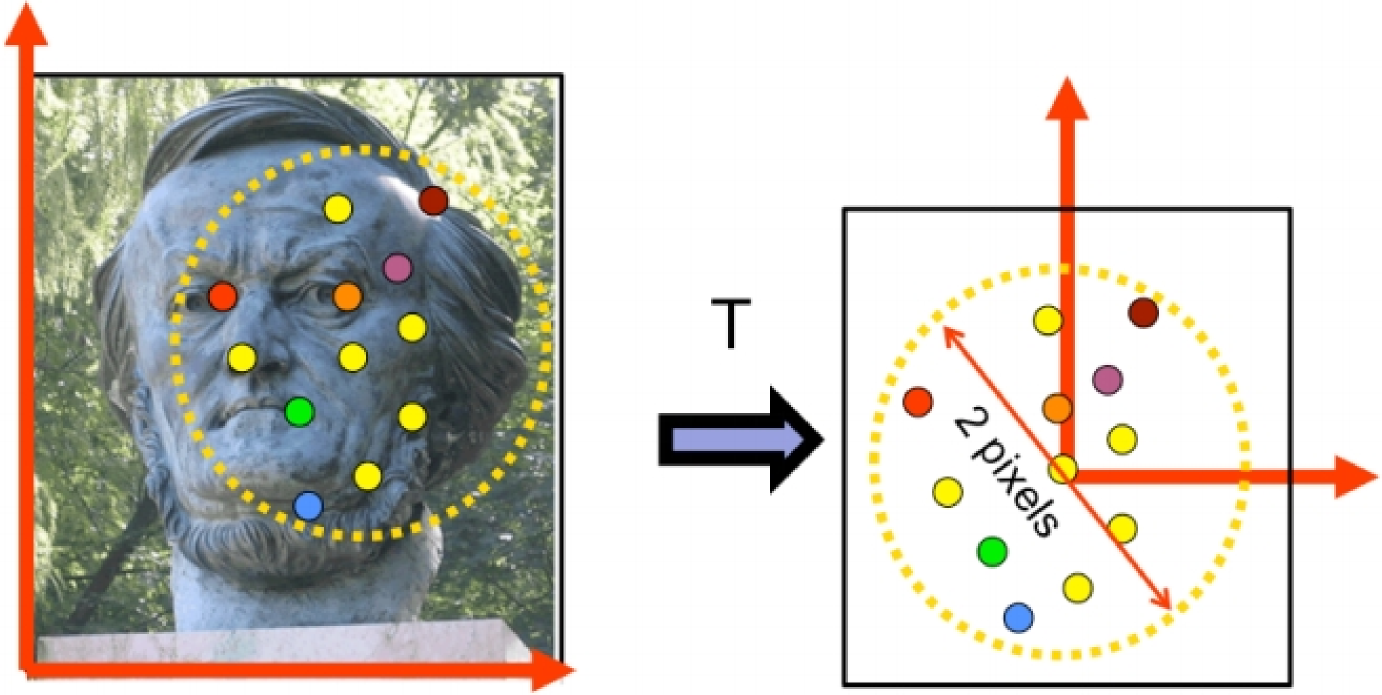
- Solution $\longrightarrow$ normalize A
- Apply a transformation T (translation + scale) to image coordinates
  - Basically a similarity without a rotation
- We set:
  - Centroid of points as origin
  - T and S leading to average distance from the origin ~ $\sqrt{2}$ pixels

See HZ page 108 for details

Normalized 8 points algorithm



Coordinate system of the image before applying T

Coordinate system of the image after applying T

See HZ page 108 for details

Normalized 8 points algorithm

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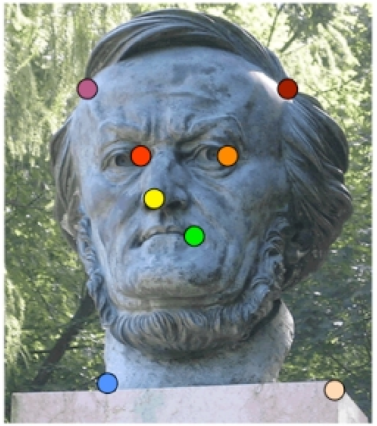
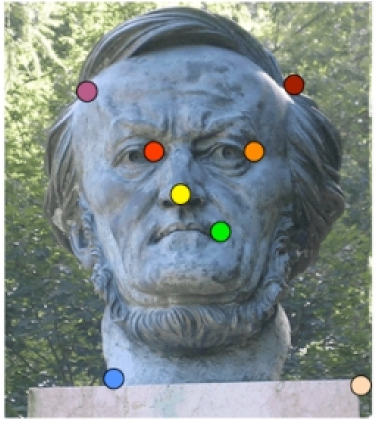
- Apply a T and T' to left and right images
- Obtain a "new" set of, at least, 8 points and proceed as before



$$q_i = T\ p_i$$



$$q_i' = T'\ p_i'$$

See HZ page 108 for details

Normalized 8 points algorithm

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- Once we obtain a fundamental matrix F_q we need to de-normalize it
- Simply invert the transformation

$$\begin{cases} q^T F_q q' = 0 \\ det(F_q) = 0 \end{cases} \qquad F = T^T F_q T'$$

See HZ page 108 for details

Recap

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- Compute T & T' for right/left images
- Normalize the coords as $q_i = Tp_i$ and $q'_i = Tp'_i$
- Apply the 8 points approach to compute a $\hat{F}_q$
- Enforce rank-2 constraint to obtain F_q
- Once we obtain a fundamental matrix F_q we need to de-normalize it, inverting T & T' transormations

See HZ page 108 for details

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8 Points Algorithm

Question time!

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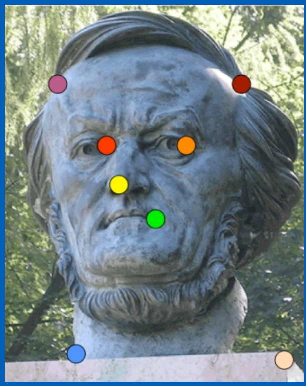
8 Points Algorithm



Algoritmo degli 8 punti

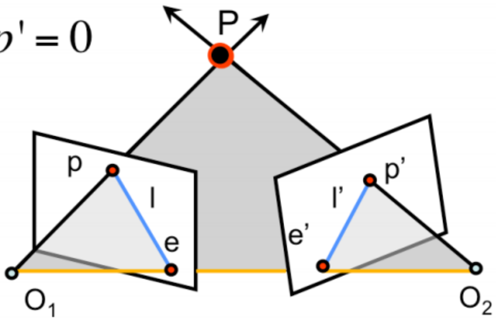
Summary

- Recap epipolar constraints
- 8 points method
- Normalized 8-points method

Un piccolo recap

Epipolar constraints

$$p^T \cdot F\, p' = 0$$



- l = Fp' ⟶ epipolar line for p
- l' = F^Tp ⟶ epipolar line for p'
- F is a 3x3 matrix with 7 DOF
- F is singular and its rank is 2

Limitiamoci alla matrice fondamentale, che poi è anche il caso piú complesso. Ovvero quello di cui non so niente delle mie dure telecamere.

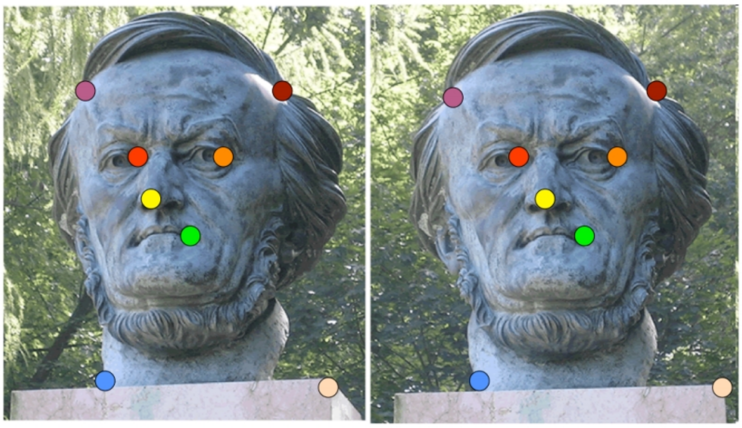
Se trovo la F riesco a identificare la linea epipolare su cui cercare le corrispondenze dall'altro lato. O meglio la userò per rettificare le immagini.

Perché 7 e non 9? Le coordinate omogenee fanno sì che la scala mi arrivi nel risultato e il det(F)= 0 mi riduce di 1 altro vincolo.

Fundamental Matrix

- It is not necessary to fully know their calibration parameters of a stereo vision system
- A reduced set of relations between the two views suffices
- The fundamental matrix F encodes this set
  - F can be used to relate the projections of the same world point P on both images (p & p')
  - F can be used to determine epipolar lines then easing the search of correspondent points in both images
  - It can be also used to rectify stereo images

Fundamental Matrix estimation



- Can we estimate F without knowing nothing about K, R, T?
- Yes ⟶ 8-point algorithm

C'è il modo di calcolarsi la F? Sì ⟶ metodo degli 8 punti

Che poi in realtà sono 8 coppie di punti "corrispondenti" sulle due immagini.

Fundamental Matrix Estimation

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- Problem:
  - Given a set of matched image points in both images $\{p_i, p'_i\}$
  - Estimate the Fundamental Matrix F
- How many matches do we need?
- How we perform the estimation?

Formalizziamo il problema. Conosco diverse coppie di punti p e p'
Quante ne servono e comunque come calcolo la F?

Expand equation

- Each correspondence should satisfy the epipolar constraint

$$p_i^T F\, p_i' = 0$$

$$p_i = \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} \qquad p_i' = \begin{bmatrix} u_i' \\ v_i' \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} u_i & v_i & 1 \end{bmatrix} \begin{bmatrix} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{bmatrix} \begin{bmatrix} u_i' \\ v_i' \\ 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} \boldsymbol{u_i u_i'} & \boldsymbol{u_i v_i'} & \boldsymbol{u_i} & \boldsymbol{v_i u_i'} & \boldsymbol{v_i v_i'} & \boldsymbol{v_i} & \boldsymbol{u_i'} & \boldsymbol{v_i'} & \boldsymbol{1} \end{bmatrix} \begin{bmatrix} \boldsymbol{F_{11}} \\ \boldsymbol{F_{12}} \\ \boldsymbol{F_{13}} \\ \boldsymbol{F_{21}} \\ \boldsymbol{F_{22}} \\ \boldsymbol{F_{23}} \\ \boldsymbol{F_{31}} \\ \boldsymbol{F_{32}} \\ \boldsymbol{F_{33}} \end{bmatrix} = \boldsymbol{0}$$

Data una singola coppia vale quella relazione. Cerchiamo di scriverla in modo opportuno. Rammento che I vari u e v sono noti, le incognite sono gli F

Linear Homogeneous System

- One correspondence ⟶ One equation
- Multiple correspondences ⟶ **Linear Homogenoeous System**

$$\begin{bmatrix} u_1 u_1' & u_1 v_1' & u_1 & v_1 u_1' & v_1 v_1' & v_1 & u_1' & v_1' & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_i u_i' & u_i v_i' & u_i & v_i u_i' & v_i v_i' & v_i & u_i' & v_i' & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_n u_n' & u_n v_n' & u_n & v_n u_n' & v_n v_n' & v_n & u_n' & v_n' & 1 \end{bmatrix} \begin{matrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{matrix} = 0$$

È un po' diverso il caso visto per la calibrazione. Qui 1 relazione ci fornisce 1 equazione.

Correspondences

- Therefore 1 correspondence $\longrightarrow$ 1 equation
- 9 unknowns $\longrightarrow$ 9 equations?
  - No, fundamental matrix has 7 DOF
- Therefore 7 correspondences should be enough
  - Yes, if we would like to explore complex non linear methods
  - We definitely would opt for a fast linear solution
  - Then use 8 points!

Se abbiamo che nella nostra matrice ci sono 7 gradi di libertà basterebbero 7 coppie. Con 8 però si semplifica il problema. Hartman stesso (pag. 281) ci dice "The 8-point algorithm is the simplest method of computing the fundamental matrix, involving no more than the construction and (least-squares) solution of a set of linear equations."

Correspondences

- Take then, at least, 8 correspondences

$$
\begin{bmatrix}
u_1u'_1 & u_1v'_1 & u_1 & v_1u'_1 & v_1v'_1 & v_1 & u'_1 & v'_1 & 1 \\
u_2u'_2 & u_2v'_2 & u_2 & v_2u'_2 & v_2v'_2 & v_2 & u'_2 & v'_2 & 1 \\
u_3u'_3 & u_3v'_3 & u_3 & v_3u'_3 & v_3v'_3 & v_3 & u'_3 & v'_3 & 1 \\
u_4u'_4 & u_4v'_4 & u_4 & v_4u'_4 & v_4v'_4 & v_4 & u'_4 & v'_4 & 1 \\
u_5u'_5 & u_5v'_5 & u_5 & v_5u'_5 & v_5v'_5 & v_5 & u'_5 & v'_5 & 1 \\
u_6u'_6 & u_6v'_6 & u_6 & v_6u'_6 & v_6v'_6 & v_6 & u'_6 & v'_6 & 1 \\
u_7u'_7 & u_7v'_7 & u_7 & v_7u'_7 & v_7v'_7 & v_7 & u'_7 & v'_7 & 1 \\
u_8u'_8 & u_8v'_8 & u_8 & v_8u'_8 & v_8v'_8 & v_8 & u'_8 & v'_8 & 1
\end{bmatrix}
\begin{bmatrix}
F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33}
\end{bmatrix}
= Af = 0
$$

Questo è un sistema omogeneo $\longrightarrow$ lineare
Sappiamo già come risolverlo

Fundamental Matrix Estimation

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- Homogeneous system
- F_{ij}s are the unknowns
  - We can find them up to a scale → impose norm as 1
- A not 0 solution is possible
  - For 8 points when rank of A is 8
  - Noise and quantization can lead to have 9 as rank (9 columns)

Solito sistema omogeneo, come sempre risolvibile al netto di una scala ed esiste la soluzione F=0 che dobbiamo però scartare

Fundamental Matrix Estimation

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- Again, total least squares approach
  - $Af = 0 \longrightarrow$ minimize $||Af||^2$
- SVD can be used
  - In order to avoid the trivial zero solution and bypass scale issue impose the norm to be 1
  - $||F||^2 = 1$

Usiamo lo stesso approccio della calibrazione e per risolvere il problema scala qui possiamo imporre norma = 1

Fundamental Matrix Estimation

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- Find the SVD of $A^T A$
- Entries of f (or F) are the column of V corresponding to the least singular value

Fundamental Matrix Estimation

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- Solving previous system gives us a $\hat{F}$ solution that satisfies
  - $p^T\hat{F}p'=0$
- Anyway $\hat{F}$ is not necessarily a proper fundamental matrix
  - i.e. it can not be a rank 2 matrix
- Let's look for a best rank 2 approximation
  - We seek for an F that minimizes the Froebenius norm of $\hat{F}$
  - Subject to the constraint that det(F) = 0

La soluzione che troviamo non è detto sia quella definitiva anzi probasbilmente non lo è causa le approssimazioni in gioco. Tipicamente il rango della F trovata non è pari a 2

Fundamental Matrix Estimation

- Therefore we have to solve
  - $||F - \hat{F}|| = 0$ (Froebenius norm)
  - $\det(F) = 0$
- Froebenius norm of a matrix:
  - Sq. root of sum of squares of all entries of the matrix
- How to solve? $\longrightarrow$ SVD again

Aggiungiamo questi due requisiti e risolviamo nuovamente con SVD
Esisterebbe un approccio migliore che prevede l'uso della distanza di Mahalanobis però è complesso

Fundamental Matrix Estimation

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$$\|F-\hat{F}\|=0 \qquad det(F)=0$$

$$F=U\begin{bmatrix} s_1 & 0 & 0 \\ 0 & s_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}V^T \qquad UDV^T=SVD(\hat{F})$$

[HZ] pag 281, chapter 11, “Computation of F”

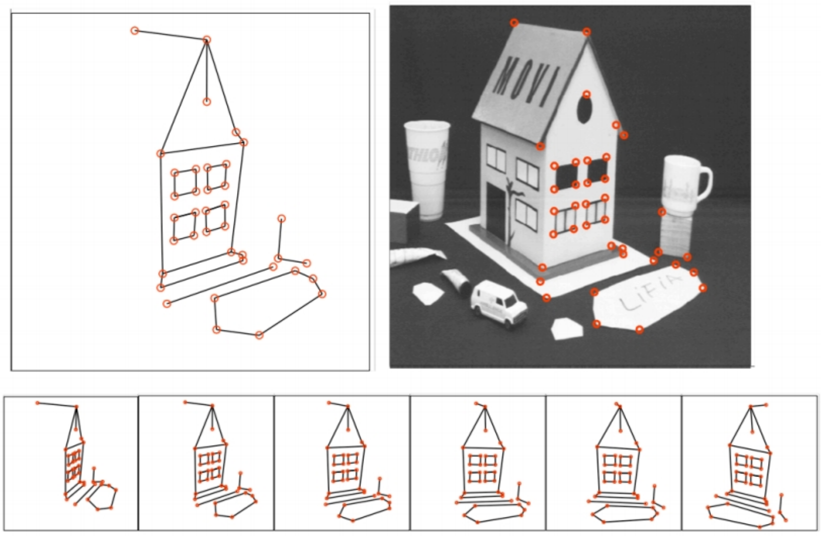
Si noti come uno dei valori singolari DEVE essere 0 e s1 e s2 dovrebbero essere uguali

Fundamental Matrix Estimation

- The 8-point algorithm may be then formulated as consisting of two steps
  1) Linear solution. A solution $\hat{F}$ is obtained from the vector f corresponding to the smallest singular value of A
  2) Constraint enforcement. Replace $\hat{F}$ by F , the closest singular matrix to $\hat{F}$ under a Frobenius norm. This correction is done using the SVD
- The algorithm thus stated is extremely simple, and readily implemented

L'ultima frase viene dall'Hartley...

Slection of points



Data courtesy of R. Mohr and B. Boufama.

Tipicamente la selezione delle corrispondenze è comunque automatica

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Normalized 8 Points Algorithm

Però scegliere gli 8 punti “a caso” può portare a soluzioni problematiche

Normalized 8 points algorithm

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- The 8-point algorithm features a numerical issue $Af = 0$
- Matrix A can be a not well conditioned (unbalanced) matrix
  - Values of matrix A are not necessarily in the same order of magnitude
  - This can badly affect the SVD decomposition
- No invariance wrt similarity ([SR T])

See HZ page 108 for details

A può contenere vaslori significativamente differenti tra di loro se i punti non sono normalizzati. Ricordiamoci che l'uso di SVD è comunque un approssimazione.
Quindi potenziale problema dal punto di vista numerico.
Il problema è invariante rispetto alla similarità? → NO

Normalized 8 points algorithm

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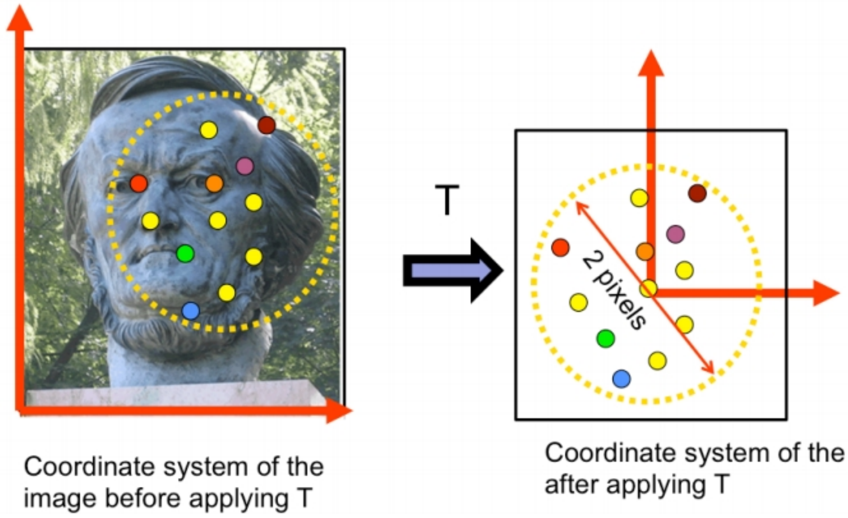
- Solution $\longrightarrow$ normalize A
- Apply a transformation T (translation + scale) to image coordinates
  - Basically a similarity without a rotation
- We set:
  - Centroid of points as origin
  - T and S leading to average distance from the origin ~ $\sqrt{2}$ pixels

See HZ page 108 for details

L'idea di base è normalizzare A.
Lo faccio spostando l'origine sul centroide dei miei punti e scalando di modo che I numeri in oggetto soddisfino quel limite

Normalized 8 points algorithm

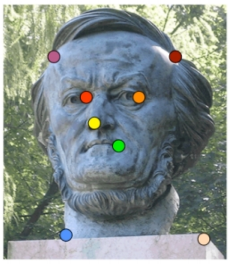


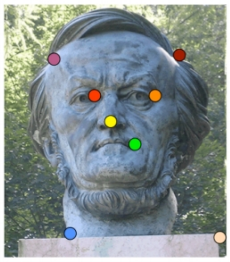
See HZ page 108 for details

Le distanze medie al quadrato diventano circa 2 pixel

Normalized 8 points algorithm

- Apply a T and T' to left and right images
- Obtain a "new" set of, at least, 8 points and proceed as before





$q_i = T\ p_i$ $\qquad$ $q_i' = T'\ p_i'$

See HZ page 108 for details

Normalized 8 points algorithm

- Once we obtain a fundamental matrix F_q we need to de-normalize it
- Simply invert the transformation

$$\begin{cases} q^T F_q q' = 0 \\ det(F_q) = 0 \end{cases} \qquad F = T^T F_q T'$$

See HZ page 108 for details

Il sistema è per ottenere Fq da $\overline{Fq}$

Recap

- Compute T & T' for right/left images
- Normalize the coords as $q_i=Tp_i$ and $q'_i=Tp'_i$
- Apply the 8 points approach to compute a $\hat{F}_q$
- Enforce rank-2 constraint to obtain F_q
- Once we obtain a fundamental matrix F_q we need to de-normalize it, inverting T & T' transormations

See HZ page 108 for details

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8 Points Algorithm

Question time!

Però scegliere gli 8 punti “a caso” può portare a soluzioni problematiche